Time Series Analysis and Forecasting

Seminar 13: LPPL Models for Bubble Detection

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Seminar Outline

- **Part I: Year 3 Level** – Super-exponential growth, LPPL equation, model fitting
- **Part II: Master Level** – Filter conditions, benchmarks, bootstrap CI, multi-window stability, LPPLS CI
- **Part III: PhD Extension** – Negative control (COVID 2020), risk management application
- **Discussion Topics** – Critical thinking with AI tools
- **Take-Home Exercises** – Extended practice
- **Summary** – Key concepts by level
- **Bibliography** – References and code

Exercise 1: Super-Exponential Growth Detection

Question

Download BTC-USD daily prices from Sep 2020 to Nov 2021 (the run-up to \$69k). Test whether the log-price series exhibits **super-exponential growth**.

Exercise

- **(A)** Plot price, log-price, daily returns, and 30-day rolling volatility (4 panels).
- **(B)** Fit three models to $\ln P_t$: linear ($a + bt$), quadratic ($a + bt + ct^2$), and exponential ($a + be^{ct}$). Report R^2 for each.
- **(C)** If $R^2_{\text{quadratic}} > R^2_{\text{linear}}$, what does this imply about the growth regime? Why is this a necessary condition for LPPL applicability?

Answer on next slide...

Exercise 1: Solution

Solution

- **(A)** Price: exponential-looking surge from \$10k to \$69k. Log-price: concave upward (faster-than-linear). Returns: heteroskedastic with increasing amplitude. Rolling vol: rising trend toward the peak.
- **(B)** Typical results: $R^2_{\text{lin}} \approx 0.92$, $R^2_{\text{quad}} \approx 0.97$, $R^2_{\text{exp}} \approx 0.98$. The nonlinear models substantially outperform the linear fit.
- **(C)** If log-price grows faster than linearly, then price grows faster than exponentially — this is **super-exponential growth**. It arises from positive feedback loops (herding, momentum trading) and is the hallmark of a bubble regime. LPPL models specifically describe this regime: the power-law singularity $B(t_c - t)^m$ with $m < 1$ produces super-exponential acceleration.

TSA_ch13_btc2021_data

Exercise 2: The LPPL Equation

Question

The LPPL model describes log-price during a bubble:
$\ln P(t) = A + B(t_c - t)^m + C(t_c - t)^m \cos\big(\omega \ln(t_c - t) - \varphi\big)$

Exercise

- ☐ **(A)** Using synthetic parameters (A=11, B=−0.5, m=0.5, C=0.04, ω=8, φ=1, t_c=300), plot the three components: power-law trend, log-periodic oscillations, full LPPL.
- ☐ **(B)** Why must $B < 0$ and $0 < m < 1$ for a bubble? What happens if $m \geq 1$?
- ☐ **(C)** Why do oscillations accelerate in calendar time but have constant frequency in $\ln(t_c - t)$?

Answer on next slide...

Exercise 2: Solution

Solution

- **(A)** Component 1: $A + B(t_c - t)^m$ curves upward as $t \to t_c$ (power-law singularity). Component 2: $C(t_c - t)^m \cos(\omega \ln(t_c - t) - \varphi)$ — oscillations that shrink in amplitude and compress in period near t_c. Combined: super-exponential trend decorated with accelerating oscillations.
- **(B)** $B < 0$ ensures the function *increases* as $t \to t_c$ (since $(t_c - t)^m$ decreases). $0 < m < 1$ ensures **super-exponential growth**: the growth rate $|\dot{P}/P|$ diverges as $t \to t_c$ (finite-time singularity). If $m \geq 1$, growth decelerates — no bubble signature.
- **(C)** The oscillation argument is $\omega \ln(t_c - t)$. Equal increments of $\ln(t_c - t)$ correspond to geometric compression in $(t_c - t)$: successive cycles take $1/\lambda$ the time of the previous one, where $\lambda = e^{2\pi/\omega} \approx 2$. This is **discrete scale invariance**.

TSA_ch13_lppl_components

Exercise 3: LPPL Fitting via Partial Linearization

Question

The LPPL has 7 parameters $(A, B, C_1, C_2, t_c, m, \omega)$ where $C_1 = C\cos\varphi$, $C_2 = C\sin\varphi$. For fixed (t_c, m, ω), the remaining parameters can be found by OLS.

Exercise

- **(A)** Write the OLS design matrix X for the linear sub-problem. Identify each column.
- **(B)** Use `scipy.optimize.differential_evolution` to search over (t_c, m, ω) and fit LPPL to the BTC 2021 data. Report estimated parameters and R^2.
- **(C)** Plot the LPPL fit against observed log-price and the residuals. Are residuals stationary?

Answer on next slide...

Exercise 3: Solution

Solution

- **(A)** For fixed (t_c, m, ω), define $\Delta_i = t_c - t_i$ and $f_i = \Delta_i^m$:

$$X = \begin{bmatrix} 1 & f_1 & f_1 \cos(\omega \ln \Delta_1) & f_1 \sin(\omega \ln \Delta_1) \\ \vdots & \vdots & \vdots & \vdots \\ 1 & f_n & f_n \cos(\omega \ln \Delta_n) & f_n \sin(\omega \ln \Delta_n) \end{bmatrix}, \quad \beta = \begin{pmatrix} A \\ B \\ C_1 \\ C_2 \end{pmatrix}$$

- **(B)** Bounds: $t_c \in [N{-}5, N{+}60]$, $m \in [0.1, 0.9]$, $\omega \in [4, 25]$. Typical BTC 2021 results: $t_c \approx N + 5$, $m \approx 0.4$, $\omega \approx 8$, $R^2 > 0.99$.
- **(C)** The fit closely tracks the observed log-price. Residuals fluctuate around zero with small magnitude; check stationarity via Ljung-Box test on residuals.

TSA_ch13_lppl_fit

Exercise 4: The Eight Filter Conditions

Question

Sornette et al. (2015) propose 8 filter conditions that LPPL parameters must satisfy for a valid bubble signal.

Exercise

- **(A)** List the 8 conditions and explain the economic rationale behind each.
- **(B)** Check all 8 conditions on your BTC 2021 LPPL fit. Display pass/fail for each.
- **(C)** Why is the condition $B < 0$ essential? What would $B > 0$ mean economically?

Answer on next slide...

Exercise 4: Solution

Solution

- **(A)** The 8 conditions:
  1. $B < 0$: super-exponential growth (price rises as $t \to t_c$)
  2. $0.1 \leq m \leq 0.9$: power-law exponent in empirical range
  3. $4 \leq \omega \leq 25$: log-frequency in observed range
  4. $t_c > N$: critical time after the last observation
  5. $t_c < N + 0.2N$: critical time within a reasonable horizon
  6. $|C| > 0$: log-periodic oscillations are present
  7. $m|B| > 0$: finite-time singularity has detectable strength
  8. $\omega/(2\pi) \geq 1$: at least one full oscillation cycle
- **(B)** For BTC 2021, typically 8/8 conditions pass.
- **(C)** $B > 0$ would mean price *decreases* as $t \to t_c$ — this describes an **anti-bubble** (deflationary spiral), not a speculative bubble.

TSA_ch13_filter_conditions

Exercise 5: Benchmark Models

Question

A 7-parameter model like LPPL can overfit. Compare it against simpler alternatives to justify its use.

Exercise

- **(A)** Fit linear ($k = 2$), quadratic ($k = 3$), exponential ($k = 3$), and power-law without oscillations ($k = 4$) models to the same BTC 2021 log-price data.
- **(B)** Compute R^2, AIC, and BIC for all models including LPPL ($k = 7$). Present in a comparison table.
- **(C)** Is the LPPL improvement over the power-law model (which has the same trend but no oscillations) statistically significant? What does this tell us about the role of log-periodicity?

Answer on next slide...

Exercise 5: Solution

Solution

- **(A)** Power-law (no oscillations): $\ln P(t) = A + B(t_c - t)^m$ with 4 parameters. Fit using differential evolution for (t_c, m) and OLS for (A, B).
- **(B)** LPPL typically achieves the lowest AIC/BIC despite having 7 parameters, because the log-periodic oscillations capture structure missed by simpler models. The AIC penalizes extra parameters, so LPPL must provide a substantial fit improvement to overcome this penalty.
- **(C)** The LPPL vs. power-law comparison isolates the contribution of the $\cos(\omega \ln(t_c - t))$ term. If AIC(LPPL) $<$ AIC(power-law), the log-periodic oscillations carry genuine predictive information about the timing of the crash — they are not fitting noise.

TSA_ch13_btc2021_fit

Exercise 6: Bootstrap Confidence Intervals

Question

The point estimate of t_c is useless without an uncertainty measure. Implement residual resampling bootstrap to construct 95% CIs.

Exercise

- **(A)** Describe the residual bootstrap algorithm for LPPL: (1) fit original data, (2) compute residuals, (3) resample residuals with replacement, (4) create synthetic series, (5) re-fit. Repeat $B = 200$ times.
- **(B)** Plot histograms of t_c, m, and ω from the bootstrap. Report 95% CIs. Convert the t_c CI to calendar dates.
- **(C)** Create a scatter plot of m vs. ω from the bootstrap. What does the shape of the joint distribution tell us about parameter identifiability?

Answer on next slide...

Exercise 6: Solution

Solution

- **(A)** The residual bootstrap preserves the LPPL structure: $y_t^* = \hat{y}_t + \varepsilon_t^*$ where ε_t^* are drawn with replacement from the original residuals $\hat{\varepsilon}_t = y_t - \hat{y}_t$. Each bootstrap sample is re-fitted from scratch using differential evolution.
- **(B)** For BTC 2021, a typical 95% CI for t_c: approximately $\pm$5–10 trading days around the point estimate. m: typically tight CI $\in [0.3, 0.6]$. ω: moderate spread reflecting the challenge of pinpointing oscillation frequency.
- **(C)** The m–ω scatter often shows a negative correlation: higher m values pair with lower ω. A tight ellipse indicates good identifiability; a diffuse cloud suggests the parameters are poorly identified (multiple local optima).

TSA_ch13_bootstrap_ci

Exercise 7: Multi-Window Stability

Question

A robust bubble signal should be stable across different fitting window lengths.

Exercise

- **(A)** Fit LPPL over windows of $\{60, 90, 120, 150, 180, 200, 250, 300\}$ trading days, all ending at the same date. For each window, check all 8 filter conditions.
- **(B)** Plot t_c estimates vs. window length. Color-code points by pass/fail status. Is t_c stable across windows?
- **(C)** Plot m and ω across windows on a dual-axis chart. Do the parameters converge as the window grows?

Answer on next slide...

Exercise 7: Solution

Solution

- **(A)** Each window produces an independent LPPL fit. Windows that fail one or more filter conditions are flagged. For BTC 2021, most windows with ≥ 90 days should pass all 8 filters.
- **(B)** If t_c estimates cluster tightly (say within ± 10 days), the bubble signal is **robust**. Scattered t_c estimates across windows indicate instability — the signal may not be reliable.
- **(C)** Convergence of m and ω across windows strengthens the case that the LPPL model captures a genuine underlying dynamical process rather than fitting noise in a particular sample window.

TSA_ch13_confidence_indicator

Exercise 8: LPPLS Confidence Indicator

Question

The LPPLS Confidence Indicator (CI) provides a real-time summary of bubble risk at each date.

Exercise

- **(A)** Define the CI: at each date t, fit LPPL over multiple windows ending at t, compute $\mathrm{CI}(t) = \frac{\#\text{ valid fits}}{\#\text{ total windows}}$. Use windows $\{60, 90, 120, 150, 180\}$.
- **(B)** Compute CI every 5 trading days for BTC 2021. Plot price and CI with traffic-light coloring: green (< 0.3), amber (0.3–0.6), red (≥ 0.6).
- **(C)** Does the CI rise before the peak price, providing an early warning signal?

Answer on next slide...

Exercise 8: Solution

Solution

- **(A)** At each evaluation date, fit LPPL on 5 overlapping windows. A fit is "valid" if all 8 filter conditions pass. $\mathrm{CI}(t) \in [0, 1]$ is the fraction of valid fits. Higher CI = stronger bubble signal.
- **(B)** The traffic-light classification provides an intuitive risk dashboard:
  - **Green** ($\mathrm{CI} < 0.3$): no significant bubble signal
  - **Amber** ($0.3 \leq \mathrm{CI} < 0.6$): emerging bubble signal — increase monitoring
  - **Red** ($\mathrm{CI} \geq 0.6$): strong bubble warning — consider reducing exposure
- **(C)** In the BTC 2021 case, the CI typically turns red several weeks before the peak, providing actionable early warning.

TSA_ch13_confidence_indicator

Exercise 9: Negative Control — COVID 2020

Question

The S&P 500 crashed 34% in Feb–Mar 2020 due to COVID-19 — an exogenous shock, not an endogenous bubble. A valid bubble detection method should **not** produce a false alarm for such crashes.

Exercise

- **(A)** Download S&P 500 from Jan 2019 to Feb 19, 2020 (the pre-crash peak). Fit LPPL.
- **(B)** Check the 8 filter conditions. How many pass?
- **(C)** Compare the BTC 2021 fit (valid bubble) vs. the S&P pre-COVID fit side by side. Why does LPPL correctly distinguish between endogenous and exogenous crashes?

Answer on next slide...

Exercise 9: Solution

Solution

- **(A)** The S&P 500 pre-COVID period shows steady growth but not super-exponential acceleration. The LPPL fit is visually poor compared to the BTC case.
- **(B)** Typically only 3–5 out of 8 conditions pass. The key failures: B may not be significantly negative, m may fall outside $[0.1, 0.9]$, and ω may be poorly identified.
- **(C)** LPPL detects **endogenous** instability (herding, positive feedback) that builds up gradually before a crash. The COVID crash was driven by an **external** event (pandemic), not by internal market dynamics. No positive feedback loop $\Rightarrow$ no super-exponential growth $\Rightarrow$ no valid LPPL signal. This validates LPPL as a tool that distinguishes bubble-driven crashes from exogenous shocks.

TSA_ch13_covid2020_case

Exercise 10: Risk Management Application

Question

Use the LPPLS Confidence Indicator to design a dynamic position-sizing strategy.

Exercise

- **(A)** Define a position-sizing rule: $w(t) = \max\left(0.1,\ 1 - 1.5 \cdot \mathrm{CI}(t)^{1.5}\right)$. Plot the position size over time alongside the BTC price.
- **(B)** Compare buy-and-hold vs. the LPPL-adjusted strategy: compute cumulative returns, max drawdown, annualized Sharpe ratio, and daily volatility.
- **(C)** What are the limitations of using LPPL for real-time risk management? How would transaction costs and look-ahead bias affect the strategy?

Answer on next slide...

Exercise 10: Solution

Solution

- **(A)** The position shrinks smoothly from 100% to 10% as CI rises. This avoids binary "all-in / all-out" decisions and provides proportional exposure reduction.
- **(B)** The LPPL-adjusted strategy typically achieves: lower maximum drawdown, lower daily volatility, and higher risk-adjusted returns (Sharpe ratio) compared to buy-and-hold, at the cost of slightly lower total return.
- **(C)** Limitations: (i) parameter estimation delay — each LPPL fit requires > 60 days of data, so the CI lags real-time price action; (ii) transaction costs from position changes can erode returns; (iii) look-ahead bias if filter conditions reference the in-sample t_c; (iv) false positives — some windows may produce spurious bubble signals; (v) LPPL cannot predict the exact crash date, only the approximate critical time window.

TSA_ch13_risk_management

Discussion Topics

Topic 1: EMH vs. LPPL

- EMH says crashes are unpredictable random events. LPPL says some crashes are preceded by detectable super-exponential growth. Can both be partially correct?

Topic 2: Overfitting Risk

- LPPL has 7 parameters fitted to sometimes < 100 observations. How do filter conditions, AIC/BIC, and bootstrap protect against overfitting?
- Would you trust an LPPL signal for a real trading decision?

Topic 3: Ethical Dimensions

- If a central bank detects a bubble using LPPL, should it act preemptively?
- If a hedge fund uses LPPL to time a market exit, could this become self-defeating if widely adopted?

AI-Assisted Exercise: Critical Thinking

Exercise: Evaluating AI-Generated LPPL Analysis

- **(A)** Ask an AI assistant (ChatGPT, Claude, etc.) to explain the LPPL model and its parameters. Evaluate the response: are the parameter constraints correct? Does it mention the filter conditions? Does it explain partial linearization?
- **(B)** Ask the AI to write Python code to fit LPPL to a price series. Test the code: does it handle the nonlinear/linear split correctly? Does it use appropriate optimization bounds?
- **(C)** Ask the AI: "Can LPPL predict the exact date of a crash?" Critically evaluate its answer using what you learned in this chapter about t_c uncertainty and the difference between "most probable regime change" and "guaranteed crash date."

Take-Home Exercises

Exercise A: Historical Case Study

- Choose one of: Dot-Com 2000 (NASDAQ), Shanghai 2015 (SSE), Bitcoin 2017, or Oil 2008.
- Download data, fit LPPL, check all 8 filter conditions, run bootstrap ($n = 200$), and compare against benchmark models.
- Write a 2-page report with your estimated t_c and its 95% CI in calendar dates.

Exercise B: Current Market Screening

- Screen 5 major indices (S&P 500, NASDAQ, DAX, Nikkei, Shanghai) for bubble signals using the LPPLS Confidence Indicator.
- For each index, compute the current CI value and classify as green/amber/red.
- Discuss: are any current markets showing signs of speculative excess?

Summary: Key Concepts by Level

Level	Exercise	Key Concept
Year 3	Ex. 1	Super-exponential growth detection
	Ex. 2	LPPL equation: power law + log-periodic oscillations
	Ex. 3	Partial linearization: 3D nonlinear + 4D OLS
Master	Ex. 4	8 filter conditions for valid bubble signal
	Ex. 5	Benchmark comparison: AIC, BIC, R^2
	Ex. 6	Bootstrap CI for t_c, m, ω
	Ex. 7	Multi-window robustness analysis
	Ex. 8	LPPLS Confidence Indicator (real-time)
PhD	Ex. 9	Negative control: exogenous vs. endogenous crashes
	Ex. 10	Risk management: dynamic position sizing

Key tradeoff: Detection sensitivity $\longleftrightarrow$ False alarm rate — more windows and lenient filters $\Rightarrow$ earlier detection but more false positives

Questions?

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Online Resources and Code

- **Quantlet**: `https://quantlet.com` — Code repository for quantitative methods
- **Quantinar**: `https://quantinar.com` — Quantitative methods learning platform
- **GitHub TSA**: `https://github.com/danpele/TSA/tree/main/TSA_ch13` — Python code for this seminar

Thank You!

Questions?

Seminar materials are available at: https://danpele.github.io/Time-Series-Analysis/

Quantlet   Quantinar